

In this paper, we denote that $\alpha_m: \mathbb{N} \rightarrow \mathbb{R}$, ($m \in \mathbb{N}$).

Def. A set of sequences that converges to zero

$$C_0(\mathbb{N}) \stackrel{\text{def}}{=} \{ \alpha: \mathbb{N} \rightarrow \mathbb{R} \mid \lim_{n \rightarrow \infty} \alpha(n) = 0 \}$$

And define the ℓ^1 -space

$$\ell^1(\mathbb{N}) \stackrel{\text{def}}{=} \{ \alpha \in C_0(\mathbb{N}) \mid \text{the series } \sum_{n=1}^{\infty} \alpha(n) \text{ converges absolutely} \}$$

It is a normed space, with a norm

$$\| \alpha \|_1 = \sum_{n=1}^{\infty} | \alpha(n) |.$$

Thm. The normed space $\ell^1(\mathbb{N})$ is complete metric space.

Proof. Let $\{ \alpha_m : \mathbb{N} \rightarrow \mathbb{R} \}$ be a Cauchy sequence. Then, for each $n \in \mathbb{N}$,

$$| \alpha_m(n) - \alpha_k(n) | \leq \sum_{i=1}^{\infty} | \alpha_m(i) - \alpha_k(i) | = \| \alpha_m - \alpha_k \|_1, \quad \text{for any } m, k \in \mathbb{N}.$$

So, since $\{ \alpha_m \}$ is Cauchy sequence, $\{ \alpha_m(n) \}$ is also Cauchy sequence for each $n \in \mathbb{N}$.

The $\mathbb{R}$ is complete, define $\alpha(n) = \lim_{m \rightarrow \infty} \alpha_m(n)$ for each $n \in \mathbb{N}$.

To show that $\alpha \in \ell^1(\mathbb{N})$ and $\| \alpha_m - \alpha \|_1 \rightarrow 0$ as $m \rightarrow \infty$, let $\varepsilon > 0$ be given.

Since $\{ \alpha_m \}$ is Cauchy sequence, there exists $N \in \mathbb{N}$ such that

$$m, k \geq N \Rightarrow \| \alpha_m - \alpha_k \|_1 < \varepsilon.$$

Then, for each $n \in \mathbb{N}$,

$$m, k \geq N \Rightarrow \sum_{i=1}^n | \alpha_m(i) - \alpha_k(i) | \leq \| \alpha_m - \alpha_k \|_1 < \varepsilon.$$

Letting $k \rightarrow \infty$, we obtain that, for all $n \in \mathbb{N}$

$$m \geq N \Rightarrow \sum_{i=1}^n | \alpha_m(i) - \alpha(i) | \leq \varepsilon.$$

So $\| \alpha_m - \alpha \|_1 \rightarrow 0$ as $m \rightarrow \infty$.

In particular, for each $n \in \mathbb{N}$,

$$\sum_{i=1}^n | \alpha(i) | \leq \sum_{i=1}^n | \alpha_N(i) | + \sum_{i=1}^n | \alpha(i) - \alpha_N(i) | \leq \| \alpha_N \|_1 + \varepsilon, \quad \text{so } \alpha \in \ell^1(\mathbb{N}).$$

□

